Mercury

Engineering Reflections

Lessons Learned — GPU-Class PDN Test Platform

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Preface

This document is a personal engineering reflection on the Mercury project — what worked, what didn't, what surprised me, and what I would do differently. It is not a sanitised summary of wins. The value of a project like this is not just in what gets built, but in the calibration it gives you about your own thinking and assumptions. This document tries to capture that honestly.

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1. What Went Well

1.1 The Core Insight Landed Before the Layout Did

The most important engineering conclusion of the project — that inductance dominates early transient droop, not capacitance — became clear during the LTspice parametric sweeps before a single via had been placed. That sequence mattered. Because I understood the mechanism before committing to physical geometry, every major layout decision (VCORE plane sandwiching, MLCC placement hierarchy, DrMOS-to-load path minimisation) was made deliberately, not retrofitted.

In hindsight, this is the correct order of operations: simulate to understand the dominant physics, then design to address it. I did not always work this way instinctively, and Mercury was the project that locked it in.

1.2 The Stackup Decision Was Correct — And Six Layers Was the Right Call for This Revision

The six-layer stackup with VCORE on L3 sandwiched between L2 and L4 ground planes is the right call for this current class, and I stand by that decision for what Mercury is: a first-revision testbench design. It delivered everything needed — tight loop inductance through plane sandwiching, continuous return paths, and fabrication feasibility at reasonable cost.

That said, I want to be clear: six layers is not the only correct answer here, and if I were doing this project again I would seriously consider moving to eight layers. The argument for eight is not about fixing a flaw in the six-layer design — it is about unlocking additional capability that was almost within reach but not quite achievable:

- The 12 V input distribution currently occupies L1 as a copper pour, which works well but means L1 is shared between high-current power and switching-node copper. An eighth layer would allow the 12 V distribution to be moved to a dedicated internal plane, completely freeing L1 for component placement, phase routing, and switching-node management with no compromise.
- The controller-to-DrMOS routing — gate signals, current sense lines, fault signals — was almost given a clean dedicated layer at six layers, but not quite. With eight layers there is enough room to give this signal group its own plane with proper return path control, improving signal integrity meaningfully at the switching frequencies in use. This is not a marginal SI gain; at 400–500 kHz gate drive with fast edges, clean dedicated routing matters.
- General logic routing headroom increases substantially. The 3.3 V and control signals are currently squeezed onto L5 alongside logic routing. Eight layers would allow separation between the power

management bus, the MCU control signals, and any auxiliary low-power routing — reducing coupling risk and simplifying DRC compliance in a denser revision.

The cost argument for staying at six layers was valid for this revision: the incremental fabrication cost of an additional two layers on a testbench platform was not justified by the gains available at the time. But for a Rev-A or any follow-on design where SI performance and routing cleanliness are explicit goals, eight layers represents a genuinely worthwhile improvement — not an extravagance. The lesson here is that there is more than one correct stackup choice, and the right answer depends on revision goals as much as on electrical requirements.

1.3 The PMBus Debug Architecture Was Industry-Grade

The decision to include 0 Ω resistors between the MCU and the PMBus was a small detail that reflects real production thinking. It costs nothing in board area and nothing in BOM cost, but it means the board can be debugged with an external PMBus controller without the MCU fighting for bus ownership. That is exactly what commercial GPU teams do during bring-up, and it came from understanding the problem rather than copying a reference design.

1.4 The MLCC Carpet Strategy Was Sound

Using overlapping capacitor values (220 nF, 1 µF, 4.7 µF, 22 µF) rather than a single high-value bulk MLCC is the correct architectural approach. It improves robustness to parasitic variation and placement-induced inductance differences. Targeting discrete frequency bands with individual capacitor values is fragile — component tolerances and placement inductance variation can leave gaps. Overlapping coverage is more robust. I got this right from the start.

However, whether all four value tiers are actually necessary is a different question — one covered honestly in Section 3.2.

2. What I Would Do Differently

2.1 Design for Measurement From Day One

The biggest gap in this project is the absence of physical waveform measurements. The board was designed to deliver power — but not instrumented to prove it did. Differential test points at the load node optimised for oscilloscope probing were not included in the first revision. Adding them as an afterthought in Rev-A is unnecessary friction that could have been avoided.

The lesson is simple and non-negotiable for any future revision: design for measurement before you design for performance. A result you cannot measure is a result you cannot defend. Every power delivery board I design going forward will have load-node test points as a non-negotiable first-pass requirement, not an optional feature.

2.2 The Load System: Wrong Timescale, Wrong Topology

The PADDLE load system works as a current injection mechanism, but I made two significant errors in how I thought about it — one architectural, one topological — both of which I only fully understood quite late in the design process.

The architectural error was chasing the first transient edge. The target of ≤ 40 mV droop at 500 A was framed around the initial voltage dip — the first few hundred nanoseconds of a load step. What I did not fully appreciate until late in the design is how critical on-die capacitance is for that specific window. In a real GPU, the die itself provides extremely low-inductance energy storage at very high frequency, directly at the point of consumption. Without it, a PCB-level load cannot replicate the excitation timescale of a real GPU transient, no matter how fast the MOSFET switches. The first droop event at true GPU timescales is essentially inaccessible to a board-level test platform without silicon.

This means the 40 mV target, while a valid design boundary for impedance planning, was applied to a timescale the hardware could not meaningfully contest. The load transition speed was moderated to compensate — which is honest and documented — but it sidesteps the original intent rather than solving it.

> **If I were doing this again, I would not try to target the first transient edge at all. I would explicitly define a timescale that the PCB-level capacitor network can actually fight — somewhere in the range where the MLCC carpet is genuinely dominant — and design the load to stimulate that window specifically. This would give a clean, defensible result that the hardware can actually validate, rather than a target that requires hedging.**

A direct consequence of this shift in thinking is that the capacitor network would be meaningfully smaller. If you are not trying to cover the sub-100 ns window — which you cannot cover without on-die capacitance anyway — much of the high-frequency MLCC carpet becomes unnecessary. The board area, BOM cost, and layout complexity that went into the densest part of the carpet could be significantly reduced without affecting performance in the timescale actually being tested.

The topological error was using MOSFETs switched directly to ground as the load element. This has two problems. First, it is thermally unpredictable: all dissipated power goes into the MOSFET channel with no natural current limiting, and the thermal behaviour at high pulse repetition rates is difficult to control without closed-loop temperature monitoring. Second, it is electrically blunt: the load transitions are governed by gate drive speed and MOSFET transfer characteristics, which are nonlinear and not easily characterised. The result is a load that works for coarse transient injection but not for anything requiring precise amplitude control or waveform modulation.

The better approach is a MOSFET-gated resistor bank. Each load bank consists of a power resistor of known value switched by a MOSFET gate. The load current is then I = VCORE / R, which is deterministic, linear, and independent of MOSFET RDS(on) variation. Thermal dissipation is distributed across the resistor body rather than concentrated in the MOSFET channel, making sustained operation at higher duty cycles practical without exotic cooling. The load is also naturally self-limiting — the resistor value sets a hard ceiling on current regardless of gate drive conditions. This is a cleaner, more modulated, and significantly more thermally robust load architecture, and it is what I would use in any revision.

2.3 Do Not Trust the PDN Impedance Tool

Early in the design I relied heavily on the PDN impedance tool to guide capacitor selection. The tool is not wrong in a mathematical sense — it correctly computes impedance as a function of frequency for the network you give it — but it tells an extremely optimistic story about what that impedance means for actual transient behaviour, and I trusted it more than I should have.

The tool's implicit message is: more capacitance at a given frequency band lowers impedance at that band, therefore reduces droop. This is technically true in the frequency domain, and for slow-changing or sustained load events it matters. But for the first edge of a fast transient — the window that dominates the peak droop number — the dominant variable is not the impedance magnitude at any particular frequency. It is the effective series inductance between the capacitor and the load node. The PDN tool does not show you that in any meaningful way. It shows you a flat, well-behaved impedance curve and implies your board will respond accordingly.

> **The LTspice experiment that recalibrated this was simple: I removed approximately half of the total capacitor count from the model and re-ran the transient simulation. The first droop dip barely changed. The overall droop profile was modestly worse, but not in proportion to how much capacitance had been removed. The impedance tool would have predicted a much larger degradation. The reality is that once inductance is the binding constraint — which it is, essentially always, for the first edge — adding more capacitance achieves very little. You are not limited by charge storage. You are limited by how fast that charge can reach the load.**

The practical consequence is that the PDN tool is useful for identifying gross gaps in frequency coverage and for checking that you have not left an obvious impedance peak somewhere in the mid-frequency range. It is not a useful predictor of first-edge droop performance, and using it as one leads to over-designed capacitor networks that add cost, complexity, and board area without meaningfully improving the metric you actually care about. The tool should be used for what it is good at and not trusted beyond that.

2.4 Loop Compensation Was Left Entirely Open

The ISL68229 was treated as a black box that regulates voltage, and loop compensation was not formally derived. For a first-pass exploratory design this is defensible — but only barely. The VRM control loop affects droop recovery, and without characterised bandwidth and phase margin, I cannot make confident claims about recovery behaviour from simulation alone.

This is not a fundamental flaw in the design — the compensation network can be tuned without board respins — but it is an incomplete piece of work. In a production context this would not have been shipped without loop measurement. I should hold myself to the same standard.

2.5 The Simulation Gap Was Larger Than Expected

The lumped LTspice model correctly identifies sensitivity — it tells you that inductance matters more than capacitance, and it correctly ranks design variables. But the absolute numbers it produces are conservative in a way that is difficult to quantify without measurement. Distributed current spreading in copper pours, plane-to-plane coupling, and the parallel conduction paths in a real PCB all reduce effective inductance in ways the lumped model cannot capture.

The result is that I have simulation data showing relative sensitivity but no ground truth to anchor it to. That is an uncomfortable position and is the primary motivation for building measurement infrastructure into Rev-A before anything else.

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3. What Surprised Me

3.1 How Little Bulk Capacitance Actually Helps

Going into this project, I expected bulk capacitance to be the primary tool for managing transient droop. The simulation result that removing the bulk caps made almost no difference to initial droop was genuinely surprising — not because I lacked the theory to predict it, but because the magnitude of the effect was larger than I expected. You can read that inductance dominates in a textbook and still be surprised by how aggressively it dominates in practice.

This recalibrated how I think about PDN design. Capacitance buys you recovery time. Geometry buys you initial droop performance. They are solving different parts of the same problem, and confusing them leads to designs that look over-specified on paper but underperform at the moment that matters most. The reason on-die capacitance is so significant in real GPU systems is precisely this: it does not just add more capacitance — it eliminates the inductance problem entirely by placing the energy source at zero loop inductance from the load.

3.2 How Little the Smallest Capacitors Actually Contribute

I expected the 220 nF capacitors at the innermost ring of the MLCC carpet to be the most valuable components in the network — the fastest responders, the first line of defence against the initial droop edge. What the LTspice sweeps showed was more nuanced and, honestly, somewhat deflating.

The 220 nF capacitors do respond marginally earlier than the 1 µF devices — by a small amount of time set by their self-resonant frequency and placement inductance. But they stop contributing almost immediately after that initial window. Their energy reservoir is tiny relative to the current demand, and they are essentially depleted within the first few nanoseconds of a transient. The 1 µF capacitors, by contrast, contribute for meaningfully longer before the 4.7 µF devices take over, and the transition between those tiers is visible and consequential in simulation.

> **The practical upshot: the difference in first-edge response between a board with 220 nF capacitors and a board without them is small enough that I would seriously consider removing the 220 nF tier entirely in a revised design. The board area, placement density, and assembly complexity they consume does not justify the marginal benefit they provide. The 1 µF tier is the real workhorse at high frequency; everything above that handles the mid-frequency and recovery regime. The 220 nF devices occupy the outermost edge of a window that the hardware arguably cannot address anyway without on-die capacitance.**

This finding also reinforces the earlier point about timescale targeting. If the design is explicitly not trying to win the sub-100 ns window, then the entire rationale for the 220 nF tier disappears. It was there to fight

for a transient window the board cannot access. Removing it is the honest consequence of accepting that constraint.

3.3 How Complex Via Strategy Actually Is

I knew vias mattered. I did not fully appreciate that via allocation is itself a first-order design variable with its own tradeoffs — not just a mechanical necessity for connecting layers. The tradeoff between via count (lower inductance per transition) and plane perforation (reduced current spreading, potentially higher inductance in the surrounding plane) is genuinely non-trivial. Getting it wrong in one direction increases inductance at the via itself; getting it wrong in the other direction increases inductance in the plane around it.

There is no closed-form solution to this. It requires judgment, iteration, and ideally a field solver to validate. A Rev-A with field-solver analysis of the via farm in the load region would be a meaningful improvement over the rule-of-thumb approach used in this revision.

3.4 How Much the Documentation Process Taught Me

Writing the Master Technical Report forced me to articulate design decisions I had made intuitively during layout and to defend them in prose. Several times this process revealed that a decision I thought I had made deliberately was actually made by default — I had not consciously chosen it, I had just not considered the alternative. Documentation is not just a communication tool; it is a design review tool. I will not treat it as an afterthought again.

4. What I Would Tell Someone Starting This Project

4.1 Simulate Before You Place

Build the LTspice model before you open the layout editor. Not as a final answer, but to identify the dominant physics. Once you know what dominates — in this case, inductance — every subsequent decision has a clear criterion. Without that, you are optimising by instinct and hoping for the best.

4.2 Design for Measurement First

Add differential test points at the load node, at the VRM output, and at the phase output before anything else. Everything else on the board can be debated and iterated. Without measurement infrastructure you have no ground truth and no way to validate anything you built.

4.3 The Stackup Is Important — But It Is Not a One-Time Decision

The stackup is the most consequential single decision in a high-current PDN design. Everything else — capacitor values, phase count, switching frequency — is second-order relative to the stackup. Getting it

right means the physics works for you. Getting it wrong means you spend the rest of the project fighting against it.

But here is what I would add, having been through it: there is more than one correct stackup choice for a given design, and the right answer depends heavily on revision goals, not just electrical requirements. For Mercury Rev-1, six layers was the right call — it hit the electrical targets at a fabrication cost appropriate for a testbench. For Rev-A or a more capable follow-on, eight layers is the right call — the additional routing flexibility and SI improvement justify the cost in that context.

The important thing is not to treat the stackup as irreversible once chosen. If, partway through a design, you realise that the benefits of adding two layers outweigh the cost and schedule impact of changing — add them. I stayed with six layers for Mercury partly out of inertia once the design was underway. If I were doing it again I would make that re-evaluation explicitly at the end of the placement phase, before routing was locked, and I would almost certainly move to eight layers at that point. The cost of changing a stackup mid-design is real but often smaller than the cost of the constraints it removes.

4.4 Do Not Let the PDN Tool Fool You

Use the PDN impedance tool for what it is good at: identifying frequency coverage gaps and checking that you have not left an obvious impedance peak in the mid-frequency range. Do not use it as a predictor of first-edge transient droop performance. It will tell you that more capacitance is always better and that your design looks excellent. Neither of those things is entirely true at the timescales that matter most.

Build the LTspice transient model instead. Remove components. Change inductance values. Watch what actually changes the droop. That is the experiment that teaches you what the board is actually sensitive to, and it will give you a much more honest picture than any impedance curve.

4.5 Be Honest in Your Documentation

Do not write a report that makes the project sound better than it is. Every experienced engineer reading it will spot the gaps you are hiding, and it will cost you more credibility than the gap itself would have. State what you did not measure. State what you assumed. State what you would change. That honesty is what separates a genuine engineering artifact from a portfolio piece.

5. Forward Look

Mercury Rev-A has a clear and bounded scope: add measurement infrastructure, characterise the load thermal envelope, derive loop compensation, and capture a transient droop waveform at the load node. None of this requires architectural changes — the fundamental design is sound. It requires making the board testable in the way it should have been from the start.

The load system redesign — moving from MOSFET-to-GND to MOSFET-gated resistor banks and explicitly targeting a timescale the PCB capacitor network can actually contest — is the other priority for

Rev-A. Combined with the removal of the 220 nF capacitor tier, this results in a simpler, more testable, and more thermally robust platform than the current revision.

Beyond Rev-A, the Mercury platform feeds directly into the AI accelerator chip project, where the PDN and physical design challenges are structurally similar but at smaller geometry and with tighter power domain isolation requirements. The understanding built during Mercury — particularly around inductance-dominated transient behaviour and the importance of plane geometry — transfers directly.

The honest summary of what Mercury taught me is this: power delivery engineering at the GPU and accelerator class is not about having the right components. It is about understanding the physics well enough to make the geometry work — and being honest about where the geometry still falls short.